

## ENTHUSE COURSE

PHASE : MEA,B,C,D,F,G,H,I,L,M,N,O,P,Q,R,S,T,U & MEV

TARGET : PRE-MEDICAL 2025

Test Type : SRG-MAJOR

Test Pattern : NEET (UG)

TEST DATE : 30-01-2025

### ANSWER KEY

|    |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Q. | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  | 13  | 14  | 15  | 16  | 17  | 18  | 19  | 20  | 21  | 22  | 23  | 24  | 25  | 26  | 27  | 28  | 29  | 30  |
| A. | 1   | 2   | 2   | 2   | 1   | 2   | 2   | 1   | 4   | 4   | 4   | 3   | 1   | 2   | 4   | 1   | 2   | 2   | 4   | 1   | 3   | 1   | 2   | 1   | 1   | 1   | 2   | 3   | 2   | 4   |
| Q. | 31  | 32  | 33  | 34  | 35  | 36  | 37  | 38  | 39  | 40  | 41  | 42  | 43  | 44  | 45  | 46  | 47  | 48  | 49  | 50  | 51  | 52  | 53  | 54  | 55  | 56  | 57  | 58  | 59  | 60  |
| A. | 2   | 2   | 3   | 3   | 3   | 4   | 1   | 2   | 4   | 1   | 3   | 3   | 2   | 4   | 1   | 4   | 1   | 1   | 3   | 3   | 4   | 4   | 2   | 1   | 3   | 2   | 1   | 2   | 4   | 2   |
| Q. | 61  | 62  | 63  | 64  | 65  | 66  | 67  | 68  | 69  | 70  | 71  | 72  | 73  | 74  | 75  | 76  | 77  | 78  | 79  | 80  | 81  | 82  | 83  | 84  | 85  | 86  | 87  | 88  | 89  | 90  |
| A. | 1   | 4   | 2   | 4   | 1   | 2   | 4   | 4   | 1   | 1   | 3   | 2   | 1   | 2   | 3   | 2   | 4   | 3   | 2   | 1   | 3   | 2   | 3   | 1   | 1   | 4   | 4   | 1   | 2   | 2   |
| Q. | 91  | 92  | 93  | 94  | 95  | 96  | 97  | 98  | 99  | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 |
| A. | 3   | 4   | 1   | 3   | 3   | 1   | 2   | 4   | 2   | 3   | 3   | 2   | 2   | 2   | 1   | 1   | 2   | 1   | 2   | 3   | 3   | 3   | 1   | 1   | 4   | 4   | 4   | 2   | 1   | 4   |
| Q. | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| A. | 3   | 1   | 4   | 4   | 1   | 2   | 2   | 2   | 3   | 2   | 4   | 2   | 2   | 2   | 2   | 1   | 3   | 2   | 4   | 2   | 4   | 1   | 3   | 2   | 3   | 2   | 4   | 4   | 1   | 2   |
| Q. | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 |
| A. | 2   | 3   | 4   | 2   | 1   | 4   | 2   | 1   | 1   | 2   | 2   | 4   | 2   | 2   | 2   | 1   | 1   | 4   | 4   | 1   | 4   | 3   | 4   | 4   | 1   | 3   | 1   | 1   | 4   | 1   |
| Q. | 181 | 182 | 183 | 184 | 185 | 186 | 187 | 188 | 189 | 190 | 191 | 192 | 193 | 194 | 195 | 196 | 197 | 198 | 199 | 200 |     |     |     |     |     |     |     |     |     |     |
| A. | 3   | 1   | 1   | 2   | 1   | 2   | 3   | 2   | 3   | 4   | 1   | 2   | 4   | 2   | 2   | 1   | 1   | 3   | 1   | 2   |     |     |     |     |     |     |     |     |     |     |

### HINT - SHEET

#### SUBJECT : PHYSICS

##### SECTION - A

#### 1. Ans (1)

$$g_{\text{earth}} = \frac{GM_e}{R_e^2} = g, M_e = \text{mass of earth}$$

$R_e \Rightarrow$  Radius of earth

$$g_{\text{planet}} = \frac{GM_p}{R_p^2} = \frac{G(2M_e)}{(2R_e)^2} = \frac{g}{2}$$

Now time period on earth for a simple pendulum is

$$T_{\text{earth}} = 2\pi \sqrt{\frac{\ell}{g}} = T$$

$$\therefore \text{on planet } T_{\text{planet}} = 2\pi \sqrt{\frac{\ell}{g/2}} = T$$

$$= \sqrt{2} T$$

#### 2. Ans (2)

For isothermal process

$$PV = \text{constant}$$

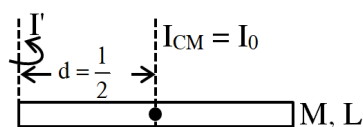
$$PdV + VdP = 0$$

$$P = -\frac{dP}{\frac{dV}{V}}$$

$$\text{Bulk modulus } B = -\frac{dP}{\frac{dV}{V}}$$

$$B_{IT} = P$$

#### 3. Ans (2)



From parallel axis theorem

$$I' = I_{CM} + Md^2$$

$$I' = I_0 + M\left(\frac{L}{2}\right)^2$$

$$I' = I_0 + \frac{ML^2}{4}$$

4. **Ans (2)**

$$L = \frac{\mu_0 N^2 A}{\ell} = \frac{\mu_0 N^2 \pi r^2}{\ell}$$

$$L \propto \frac{N^2 r^2}{\ell}$$

$$\frac{L_1}{L_2} = \left(\frac{N_1}{N_2}\right)^2 \left(\frac{r_1}{r_2}\right)^2 \left(\frac{\ell_2}{\ell_1}\right)$$

$$= \left(\frac{1}{2}\right)^2 \times \left(\frac{2}{1}\right)^2 \times \frac{1}{2} = \frac{1}{2}$$

5. **Ans (1)**

When current in wire AB increases, the magnetic flux due to outwards field linked with the loop increases. Hence, the current induced in loop must be clockwise to oppose the increase in magnetic flux.

6. **Ans (2)**

$$m_2 > m_1 \text{ so } a = \frac{m_2 - m_1}{m_2 + m_1} g = \frac{g}{4}$$

$$\Rightarrow \frac{1}{4} = \frac{m_2 - m_1}{m_2 + m_1} \Rightarrow \frac{m_1}{m_2} = \frac{3}{5}$$

7. **Ans (2)**

$$v = \frac{\omega}{k} = 3 \times 10^8$$

$$k = \frac{\omega}{v} = \frac{1.5 \times 10^7}{3 \times 10^8} = 0.05 \text{ m}^{-1}$$

9. **Ans (4)**

$$\lambda = \frac{hc}{E_{ph}} = \frac{12400}{(4\text{eV})} (\text{eV}) (\text{\AA})$$

$$= \frac{12400}{4} \text{\AA}$$

$$\lambda = 3100 \text{\AA}$$

10. **Ans (4)**

$$Q = (BE)_{\text{final}} - (BE)_{\text{initial}}$$

$$= (4)(7) - ((2 \times 1.1) + (3 \times 2.9))$$

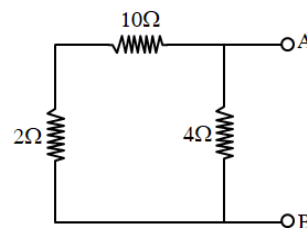
$$= 28.0 - (2.2 + 8.7)$$

$$= 28 - 10.9$$

$$= 17.1 \text{ MeV}$$

11. **Ans (4)**

Circuit can be redrawn as :



$$R_{AB} = \frac{(12)(4)}{(16)} = 3\Omega.$$

12. **Ans (3)**

$$V_d = \frac{I}{neA}, t = \frac{\ell}{V_d} = \frac{neA\ell}{I}$$

$$\Rightarrow t = \frac{(8.5 \times 10^{28})(1.6 \times 10^{-19})(2 \times 10^{-6})(5)}{4}$$

$$\Rightarrow t = \frac{(8.5)(1.6)}{4} \times 10^{28-19-6+1} = 3.4 \times 10^4 \text{ sec.}$$

13. **Ans (1)**

$$x_f - x_i = \Delta x$$

$$x_f = x_i + \Delta x = (2\hat{i} + 5\hat{j} + \hat{k}) + (8\hat{i} - 2\hat{j} + \hat{k})$$

$$= 10\hat{i} + 3\hat{j} + 2\hat{k}$$

14. **Ans (2)**

$$[\vec{\tau}] = [\vec{r} \times \vec{F}] = [ML^2T^{-2}]$$

$$[F] = [qVB]$$

$$\Rightarrow B = \left(\frac{F}{qV}\right) = \left[\frac{MLT^{-2}}{ATLT^{-1}}\right] = [MA^{-1}T^{-2}]$$

$$[M] = [I \times A] = [AL^2]$$

$$B = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2}$$

$$\Rightarrow [\mu] = \left[\frac{Br^2}{Idl}\right] = \left[\frac{MT^{-2}A^{-1} \times L^2}{AL}\right]$$

$$= [MLT^{-2}A^{-2}]$$

15. **Ans (4)**

$$g' = g - \omega^2 R \cos^2 30^\circ = 0$$

16. Ans (1)

$$E_{\text{inside}} = \frac{-GMr}{R^3} \Rightarrow E_{\text{inside}} \propto -r$$

$$E_{\text{surface}} = \frac{-GM}{R^2}$$

$$E_{\text{outside}} = \frac{-GM}{r^2} \Rightarrow E_{\text{outside}} \propto -\frac{1}{r^2}$$

17. Ans (2)

$$n_2 \lambda_2 = n_1 \lambda_1$$

$$\frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} = \frac{450}{650} = \frac{9}{13}$$

$$n_2 = 9$$

18. Ans (2)

$$r_2 = \frac{m_1 r}{m_1 + m_2} = \frac{Md}{M + 7M} = \frac{d}{8}$$

19. Ans (4)

$$A_1 V_1 = A_2 V_2$$

$$(0.1)2 = (0.05)V_2$$

$$V_2 = 4 \text{ m/s}$$

20. Ans (1)

$$\Delta P = K \frac{\Delta V}{V}$$

$$\frac{\Delta V}{V} = \frac{\Delta P}{K}$$

$$= \frac{2 \times 10^6}{4 \times 10^9} = 0.5 \times 10^{-3}$$

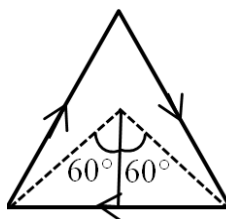
21. Ans (3)

$$B = \frac{\mu_0 N i}{\ell}$$

$$B = \frac{4\pi \times 10^{-7} \times N \times 10}{5 \times 10^{-1}} = 6.28 \times 10^{-3}$$

$$N = 250$$

22. Ans (1)



$$B_{\text{net}} = 3 B \text{ (each side)}$$

$$B_{\text{net}} = 3 \times \frac{\mu_0 I}{4\pi r} [\sin 60^\circ + \sin 60^\circ]$$

$$\left( r = \frac{a}{2\sqrt{3}} \right) \left( a = \frac{1}{3} \text{ m} \right)$$

$$= 3 \times 10^{-7} \times \frac{\sqrt{3} \times 6\sqrt{3}}{1} \times \sqrt{3} = 54 \sqrt{3} \times 10^{-7}$$

$$= 5.40 \sqrt{3} \mu\text{T}$$

23. Ans (2)

$$B = \frac{\mu_0 N i}{2r}$$

$$= \frac{4\pi \times 10^{-7} \times 100 \times 2}{2 \times 10^{-1}}$$

$$= 1.26 \times 10^{-3} \text{ Wb/m}^2$$

24. Ans (1)

$$V = \frac{kQ}{r}$$

$$Q = \frac{Vr}{k} = \frac{90 \times 20}{9 \times 10^9} = 2 \times 10^{-7} \text{ C}$$

$$= 0.2 \mu\text{C}$$

25. Ans (1)

$$\left( \vec{E}_{\text{net}} \right)_P = 0$$

26. Ans (1)

$$E_g = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{660 \times 10^{-9}} \text{ J}$$

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{660 \times 10^{-9} \times 1.6 \times 10^{-19}} \text{ eV}$$

$$= \frac{15}{8} \text{ eV}$$

$$\text{So } x = 15$$

27. Ans (2)

$$V_C = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$V_C = \frac{20 \times 500 + 10 \times 200}{30}$$

$$V_C = 400 \text{ V}$$

28. Ans (3)

$$\cos \phi = \frac{V_R}{V} = \frac{3}{\sqrt{3^2 + 4^2}} = \frac{3}{5}$$

$$\therefore \sin \phi = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \frac{4}{5}$$

$$\text{Then wattless } I \sin \phi = 2 \times \frac{4}{5} = 1.6 \text{ A}$$

29. Ans (2)

$$\Rightarrow e = \frac{(L_2 - 3L_1)}{2}$$

$$\Rightarrow e = \left( \frac{32 - 3 \times 10}{2} \right) = 1 \text{ cm}$$

30. Ans (4)

By Mechanical energy conservation

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$

$$5V^2 = 200 \times (0.2)^2$$

$$V = \sqrt{1.6} \text{ m/s}$$

31. Ans (2)

Since refraction does not depend upon intermediate

mediums, then for glass-air

$$\therefore \frac{\sin i}{\sin 90^\circ} = \frac{\mu_a}{\mu_g} \Rightarrow \sin i = \frac{1}{\mu_g} [\mu_a = 1]$$

$$\therefore \mu_g = \frac{1}{\sin i}$$

32. Ans (2)

For bi-convex lens

$$\frac{1}{f} = \frac{(\mu - 1) \times 2}{R}$$

$$\frac{1}{2f} = \frac{\mu - 1}{R} \dots (i)$$

For plano convex lens

$$\frac{1}{f'} = \frac{\mu - 1}{R} \dots (ii)$$

for (i) and (ii)

$$f' = 2f$$

33. Ans (3)

$$1 \text{ V.S.D.} = V \text{ units}$$

$$1 \text{ M.S.D.} = S \text{ units}$$

$$m \text{ V.S.D.} = (m - 1) \text{ M.S.D.}$$

$$mV = (m - 1)S$$

$$mV = mS - S$$

$$S = mS - mV$$

$$S = m(S - V)$$

$$\frac{S}{m} = S - V$$

34. Ans (3)

$$16 \ell_1 = m \ell_2$$

$$\frac{\ell_1}{\ell_2} = \frac{m}{16} \dots (1)$$

$$m \ell_1 = 4 \ell_2$$

$$\frac{\ell_1}{\ell_2} = \frac{4}{m} \dots (2)$$

On comparing (1) & (2)

$$\frac{m}{16} = \frac{4}{m}$$

$$m^2 = 64$$

$$\therefore m = 8 \text{ kg}$$

35. Ans (3)

Output of OR gate is  $y = A + B$

| A | B | $Y = A + B$ |
|---|---|-------------|
| 0 | 1 | 1           |
| 1 | 0 | 1           |
| 0 | 0 | 0           |
| 1 | 1 | 1           |

### SECTION - B

36. Ans (4)

$$PV = NK_b T$$

$$\frac{P}{K_b T} = \frac{N}{V} = n$$

$$\therefore \lambda = \frac{1}{\sqrt{2}\pi d^2 n}$$

$$\therefore \lambda = \frac{k_b T}{\sqrt{2}\pi d^2 P} \Rightarrow \lambda \propto \frac{1}{P}$$

Therefore assertion is wrong

37. Ans (1)

$$\text{Avg speed} = \frac{\sum v}{N} = \frac{1+1+1+2+3}{5} = 1.6 \text{ m/s}$$

$$\begin{aligned} \text{Avg square speed} &= \frac{\sum v^2}{N} \\ &= \frac{1^2+1^2+1^2+2^2+3^2}{5} = 3.2 \text{ m/s} \end{aligned}$$

$$v_{\text{rms}} = \sqrt{\frac{\sum v^2}{N}} = \sqrt{3.2} \approx 1.8 \text{ m/s}$$

$V_{\text{MP}} = 1 \text{ m/s}$  (speed of maximum number of molecules)

38. Ans (2)

$$E_0 = B_0 c \text{ in free space}$$

$$\vec{E} = (+10^{-6} \hat{c} \hat{j}) \cos\left(t + \frac{z}{c}\right) \frac{N}{C},$$

$$\hat{E} \times \hat{B} = \hat{V}$$

$$\hat{E} \times \hat{i} = -\hat{k}$$

$$\hat{E} = +\hat{j}$$

39. Ans (4)

Due to law of linear momentum conservation :

$$|\vec{P}_{\text{H-atom}}| = |\vec{P}_{\text{photon}}| = \frac{h}{\lambda} = Rh \left[ \frac{1}{1^2} - \frac{1}{4^2} \right]$$

$$|\vec{P}_{\text{H-atom}}| = 1.097 \times 10^7 \times 6.626 \times 10^{-34} \left[ \frac{15}{16} \right]$$

$$= 6.814 \times 10^{-27} \text{ kg-m/s}$$

40. Ans (1)

$$R = \left(10 + \frac{i}{2}\right) \Rightarrow 48 = iR = i \left(10 + \frac{i}{2}\right)$$

$$96 = 20i + i^2 \Rightarrow i^2 + 20i - 96 = 0$$

$$\Rightarrow i^2 + 24i - 4i - 96 = 0$$

$$\Rightarrow (i-4)(i+24) = 0$$

$$\Rightarrow i = 4A$$

41. Ans (3)

$$F \propto \rho^x v^y r^z \Rightarrow [F] = [\rho^x v^y r^z]$$

$$MLT^{-2} = (ML^{-3})^x (LT^{-1})^y L^z$$

Solving by comparing exponents on both sides.

$$F \propto \rho v^2 r^2$$

For the two stones  $\rho$  and  $v$  does not change.

So  $F \propto r^2$  Hence 1 : 4 is Ans.

42. Ans (3)

$$K.E_{\text{max}} = (h\nu - \phi_0)$$

(independent of intensity of incident light)

Photoelectric current increases with intensity of incident light upto saturation value.

43. Ans (2)

$$\text{By } 2\pi r = n\lambda \Rightarrow n = 3$$

$$r = n^2 r_0 = 9r_0 \Rightarrow \lambda = \frac{2\pi 9r_0}{3} = 6\pi r_0$$

for single H atom max. no. of different lines =  $n - 1$   
 $= 3 - 1 = 2$

44. Ans (4)

By CoLM

$$1 \times 5 = 1 \times v_1 + 4v_2$$

$$5 = v_1 + 4v_2$$

Here  $v_1 = 0$

$$v_2 = \frac{5}{4}$$

Loss in K.E =  $K_i - K_f$

$$= \frac{1}{2} \times 1(5)^2 - \left( \frac{1}{2} \times 1(0)^2 + \frac{1}{2} \times 4 \left( \frac{5}{4} \right)^2 \right)$$

$$= \frac{75}{8} \text{ J}$$

45. Ans (1)

$$P = \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\therefore F_2 = \frac{F_1}{A_1} \times A_2$$

$$\Rightarrow F_2 = 10F_1$$

46. Ans (4)

$$x = \sqrt{3}R$$

$$B_{\text{axis}} = \frac{\mu_0 N i R^2}{2(R^2 + x^2)^{3/2}}$$

$$= \frac{\mu_0 N i R^2}{2R^3 \times 8} = \frac{\mu_0 N i}{16R} = \frac{4\pi \times 10^{-7} \times 100 \times 3}{16\pi}$$

$$= 7.5 \times 10^{-6} \text{ T}$$

47. Ans (1)

$$F_{\text{net}} = \frac{k|x|}{1^2} + \frac{k \times 1 \times 4}{2^2} + \frac{k \times 1 \times 9}{3^2} + \dots + \frac{k \times 100}{10^2}$$

$$= 10 \text{ k N}$$

$$= 9 \times 10^{10} \text{ N}$$

48. Ans (1)

Positive cycle :  $D_1$  - forward,  $D_2$  - reverse

Negative cycle  $D_1$  - reverse,  $D_2$  - forward

49. Ans (3)

$$a = \frac{v^2}{r}$$

$\therefore v = \text{constant}$

$$\therefore a \propto \frac{1}{r}$$

$\therefore$  it is graph of rectangular hyperbola.

50. Ans (3)

$$a = -\alpha x$$

$$V \frac{dv}{dx} = -\alpha x$$

$$\int_{v_0}^0 V dv = -\alpha \int_0^x x dx$$

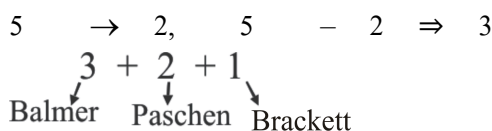
$$\frac{V_0^2}{2} = \frac{\alpha x^2}{2}$$

$$x = \frac{V_0}{\sqrt{\alpha}}$$

## SUBJECT : CHEMISTRY

### SECTION - A

51. Ans (4)



53. Ans (2)

NCERT-XI, Equilibrium Page 226 (Part-I),  
(2020-21 edition)

54. Ans (1)

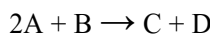
That ion hydrolyse which comes from weak acid or weak base.

55. Ans (3)

$$\text{For } 1^{\text{st}} \text{ order reaction } \frac{t_{7/8}}{t_{1/2}} = \frac{\frac{2.303}{k} \log \frac{1}{1/8}}{\frac{2.303}{k} \log \frac{1}{1/2}} = 3$$

$$t_{7/8} = 3t_{1/2}$$

56. Ans (2)



Let Initial rate (r) =  $K[A]^x [B]^y$

$$\text{Exp(1)} \quad 7.5 \times 10^{-3} = K(0.1)^x (0.1)^y \quad \dots (1)$$

$$9 \times 10^{-2} = K(0.3)^x (0.2)^y \quad \dots (2)$$

$$3.6 \times 10^{-1} = K(0.3)^x (0.4)^y \quad \dots (3)$$

$$3.6 \times 10^{-2} = K(0.4)^x (0.1)^y \quad \dots (4)$$

$$(4) \quad \frac{3.0 \times 10^{-2}}{7.5 \times 10^{-3}} = 4 = 4^x \Rightarrow \boxed{x=1}$$

$$(3) \quad \frac{3.6 \times 10^{-1}}{9 \times 10^{-2}} = 4 = 2^y \Rightarrow \boxed{y=2}$$

$$r = K(A)^1 (B)^2$$

57. Ans (1)

(3)  $\text{CHCl}_3 + \text{CH}_3\text{COCH}_3$  : -ve deviation from Raoult's law. So they form maximum boiling azeotrope.

(4)  $\text{C}_2\text{H}_5 + \text{H}_2\text{O}$  : +ve deviation from Raoult's law. So heat absorbed.

58. Ans (2)

$$\text{Approx. At. wt.} = \frac{6.4}{0.031} = 206.45$$

(लगभग परमाणु भार)

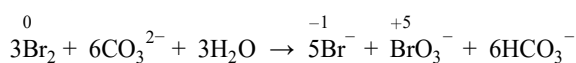
$$\text{v.f. (संयोजकता)} = \frac{\text{At. wt.}}{E} = \frac{206.45}{103.6} = 1.99$$

$$(\text{v.f.})_{\text{exact}} (\text{सही संयोजकता}) = 2$$

$$\text{Exact At. wt. (सही परमाणु भार)} = E \times \text{v.f.}$$

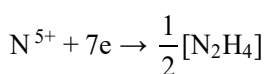
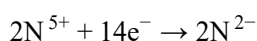
$$103.6 \times 2 = 207.2$$

59. Ans (4)



Oxidation as well as reduction of  $\text{Br}_2$  take place.

60. Ans (2)



61. Ans (1)

$$\Delta S^\circ = \sum S_{\text{P}}^\circ - \sum S_{\text{R}}^\circ = (213.8 + 197.9) - 120$$

$$= 291.7 \text{ JK}^{-1}$$

62. Ans (4)

Due to charge lyophobic colloidal particles are suspended but not coagulated.

63. Ans (2)

$$\text{IP of Na} = \text{EA of Na}^+$$

64. Ans (4)

$$\text{Ionic Radius} \propto \frac{1}{+\text{ve charge}}$$

65. Ans (1)

NCERT-XI, Part-I Pg.-94

66. Ans (2)

Due to polarisation as compound is  $\text{BeCl}_2$

67. Ans (4)

$$\text{Be}^{+2} < \text{Mg}^{+2} < \text{Ca}^{+2} < \text{Sr}^{+2} (\text{Radius})$$

Cation size  $\uparrow$ , Pol.  $\downarrow$

68. Ans (4)

NCERT-XI, Part-I Pg. 115

69. Ans (1)

$\text{AgBr}$  pale yellow, less solution in  $\text{NH}_4\text{OH}$ .

71. Ans (3)

NCERT-XII, Pg.No. 188

72. Ans (2)

$$\text{Complex is } [\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^{-1}$$

$$X = 2$$

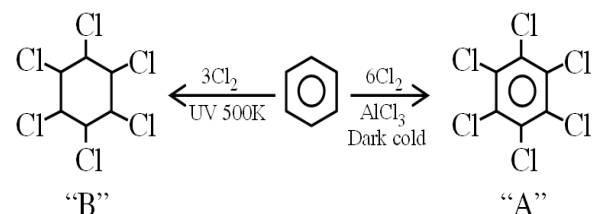
$$Y = 4$$

$$\text{So } X + Y = 6$$

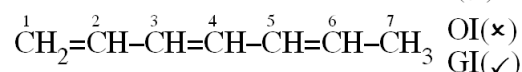
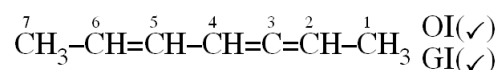
73. Ans (1)

NCERT-XII, Pg.245

75. Ans (3)

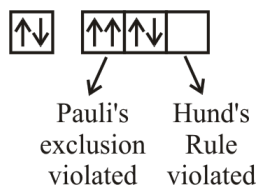


78. Ans (3)



SECTION - B

86. Ans (4)



87. Ans (4)



V.F of  $\text{H}_2\text{O} = 2$

geq. of  $\text{H}_2\text{O} = 1 \text{ mol} \times (\text{V.F} = 2)$

$= 2\text{F}$

St.-2 :- geq. of Al =  $\frac{w}{E} = \frac{40}{(27/3)} = 4.44$

$= 4.44 \text{ F}$

88. Ans (1)

$\Delta T_f = i \times k_f \times m$

$2.56 = i \times 5.12 \times 1$

$i = \frac{1}{2}$  (Association)

$\Delta T_b = i \times k_b \times m$

$2.53 = i \times 2.53 \times 1$

$i = 1$  (Normal)

89. Ans (2)

NCERT Pg. # 19

|                   |                   |                 |
|-------------------|-------------------|-----------------|
| C                 | O                 | H               |
| $\frac{57.8}{12}$ | $\frac{38.6}{16}$ | $\frac{3.6}{1}$ |
| $= 4.81$          | $= 2.4$           | $= 3.6$         |
| 2                 | 1                 | $\frac{3}{2}$   |
| 4                 | 2                 | 3               |

Empirical formula =  $\text{C}_4\text{H}_3\text{O}_2$

E.F. weight =  $4(12) + 2(16) + 3(1)$

$= 48 + 32 + 3 = 83$

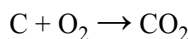
$\therefore n = \frac{\text{Molecular formula weight}}{\text{Empirical formula weight}} = \frac{166}{83} = 2$

Molecular formula =  $n[\text{E.F.}]$

$= 2[\text{C}_4\text{H}_3\text{O}_2] = [\text{C}_8\text{H}_6\text{O}_4]$

90. Ans (2)

Resonance energy =  $(\Delta H)_{\text{obs}} - (\Delta H)_{\text{Theoretical}}$



$\Delta H = (718 + 498) - (539 \times 2) = 138$

$\text{RE} = -393 - 138 = -531$

magnitude wise  $\text{RE} = 531$

91. Ans (3)

NCERT-XI, Part-I Pg.94

92. Ans (4)

NCERT-XI, Part-I Pg.131

93. Ans (1)

NCERT-XII, Pg.135

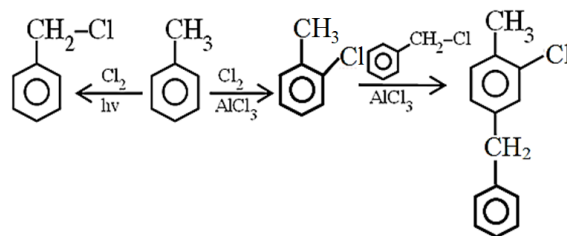
94. Ans (3)

NCERT XII, Pg. # 201

95. Ans (3)

NCERT-XII, Pg. 231

96. Ans (1)



SUBJECT : BOTANY

SECTION - A

101. Ans (3)

NCERT-XII, Pg # 22,27,30

102. Ans (2)

Module-4

103. Ans (2)

NCERT XI Page No. # 8

104. Ans (2)

NCERT-XI, Pg # 8



105. **Ans (1)**  
NCERT Pg # 35
106. **Ans (1)**  
NCERT Pg. # 36
107. **Ans (2)**  
NCERT XI Eng. Pg.# 215 (last para), 218 (2nd last para) and 220 (5th para)  
हिन्दी पेज # 216 (दूसरा पैराग्राफ)] 219 (चौथा पैराग्राफ) एवं 220 (आखिरी पैराग्राफ)
108. **Ans (1)**  
NCERT XI Pg. # 236
109. **Ans (2)**  
NCERT (XI) Pg. # 231
110. **Ans (3)**  
NCERT Pg. # 229
111. **Ans (3)**  
NCERT XI pg.# 24
112. **Ans (3)**  
NCERT Pg. # 244
113. **Ans (1)**  
NCERT-XI, Pg. # 243
114. **Ans (1)**  
NCERT XI Pg. # 61
115. **Ans (4)**  
NCERT (XIth) Pg. # 70
116. **Ans (4)**  
NCERT Pg # 191
117. **Ans (4)**  
NCERT, Pg # 197, 198
118. **Ans (2)**  
NCERT-XII, Pg # 236
119. **Ans (1)**  
NCERT (XIIth) Pg. # 220-221

120. **Ans (4)**  
NCERT Pg # 242
121. **Ans (3)**  
NCERT Pg. No. 259
123. **Ans (4)**  
NCERT XII Pg.# 96
124. **Ans (4)**  
NCERT-XII, Pg. # 60, 61
125. **Ans (1)**  
NCERT-XII, Pg. # 55,60,61
126. **Ans (2)**  
NCERT-XII, Pg. # 57
127. **Ans (2)**  
Allen module
128. **Ans (2)**  
NCERT-XII, Pg. # 78
129. **Ans (3)**  
NCERT (XII) Pg. # 103, para 4
130. **Ans (2)**  
NCERT XII, Pg # 123
131. **Ans (4)**  
NCERT-XI Pg. # 150, 151 & 152
132. **Ans (2)**  
NCERT Pg. # 89
134. **Ans (2)**  
NCERT Pg. No. 73
135. **Ans (2)**  
NCERT-XI, Pg. # 76

## SECTION - B

136. **Ans (1)**  
NCERT-XII, Pg. # 27 & 28
137. **Ans (3)**  
NCERT-XII, Pg. # 25
138. **Ans (2)**  
NCERT XI, Page No. # 6,7  
A,B & D are correct

139. **Ans ( 4 )**  
NCERT Pg # 33
140. **Ans ( 2 )**  
NCERT Pg. # 135, 136
141. **Ans ( 4 )**  
NCERT-XI, Pg. # 156
142. **Ans ( 1 )**  
NCERT (XIth) Pg. # 89
143. **Ans ( 3 )**  
NCERT (XI) Pg. # 70,71
144. **Ans ( 2 )**  
NCERT XI, Pg. # 93, para-6.3.6
145. **Ans ( 3 )**  
NCERT XII Pg. # 233
146. **Ans ( 2 )**  
NCERT-XII, # 219
147. **Ans ( 4 )**  
NCERT-XII Pg#96
148. **Ans ( 4 )**  
NCERT-XII Pg#105
149. **Ans ( 1 )**  
NCERT Pg. # 71(E), 79(H)
150. **Ans ( 2 )**  
Ncert Pg No. 74 (2022 edition) Figure 4.15

**SUBJECT : ZOOLOGY**

**SECTION - A**

151. **Ans ( 2 )**  
NCERT Pg#51
155. **Ans ( 1 )**  
NCERT Pg # 119
156. **Ans ( 4 )**  
Module

159. **Ans ( 1 )**  
NCERT Pg # 200
160. **Ans ( 2 )**  
NCERT Pg. # 193
164. **Ans ( 2 )**  
NCERT Pg. # 206
166. **Ans ( 1 )**  
NCERT Pg. # 318
168. **Ans ( 4 )**  
NCERT-XI, Pg. # 241, 242
169. **Ans ( 4 )**  
NCERT-XI, Pg. # 248
170. **Ans ( 1 )**  
NCERT-XII, Pg. # 27, Para-2.1
175. **Ans ( 1 )**  
NCERT Pg. No. # 147 [E]  
NCERT Pg. No. # 158 [H]
176. **Ans ( 3 )**  
NCERT Pg. No. # 146(E), 157(H)
179. **Ans ( 4 )**  
NCERT-XI Pg. No. # 126, 127
180. **Ans ( 1 )**  
NCERT-XI, Pg. No. # 126
182. **Ans ( 1 )**  
XI NCERT Page No. # 143 & 148
184. **Ans ( 2 )**  
NCERT Pg # 180
185. **Ans ( 1 )**  
NCERT Pg. # 153 (E), 168 (H)

**SECTION - B**

186. **Ans ( 2 )**  
NCERT-XII, Pg. # 157
187. **Ans ( 3 )**  
NCERT Pg. # 185

**188. Ans ( 2 )**

NCERT-XI, Pg. # 166

**189. Ans ( 3 )**

NCERT, Pg # 44

**192. Ans ( 2 )**

NCERT(XII) Pg# 64 (E), 71 (H)

**193. Ans ( 4 )**

NCERT Pg. # 275

**196. Ans ( 1 )**

NCERT Pg#245

**197. Ans ( 1 )**

NCERT XII Page No. # 332

**200. Ans ( 2 )**

NCERT-XII, Pg # 147

Malaria caused by Plasmodium falciparum is most harmful and may even cause death.